Matching in NC: Hall, Wronskians, and Explicit Genericity

A conceptual proof of the 2026 bipartite perfect-matching decision result

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Abstract

We give a deliberately conceptual proof that bipartite perfect-matching *decision* is in deterministic NC^2. The presentation is based on the 2026 breakthrough of Chatterjee–Ghosh–Gurjar–Raj–Thierauf and, more directly, on the subsequent simplification of Kopparty–Saraf. We discard the search and weighted versions, finite-field parameter optimization, folded Wronskians, and the formal language of subspace designs. The proof has only three ingredients: Hall's theorem, parallel matrix rank, and one Wronskian lemma saying that a low-dimensional space of low-degree polynomials has a finite "root budget."

The conceptual point is that the old randomized algorithm obtained genericity by random substitution. The new proof changes the representation: it assigns each left vertex a polynomial carrying a huge private zero, and makes each right vertex impose a high-multiplicity vanishing constraint on the sum of its neighbors. Hall deficiency then becomes a surplus of linear degrees of freedom, while Hall expansion would force a Wronskian to have more zeros than its degree permits.

The theorem to remember is not "matching has a complicated polynomial construction." It is:

Every Hall obstruction can be made visible as a kernel vector of one explicit polynomial-size matrix.

The only mechanism preventing spurious kernel vectors is the elementary principle "a nonzero polynomial cannot have too many roots," lifted from one polynomial to a low-dimensional polynomial space by a Wronskian.

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1 Why this theorem belongs in a complexity course

Let $G = (L, R, E)$ be bipartite with $|L| = |R| = n$. Hall's theorem gives the exact combinatorial obstruction:

$$G \text{ has no perfect matching} \iff \exists S \subseteq L : \ |N(S)| < |S|. \tag{1.1}$$

The problem for parallel computation is obvious: (1.1) quantifies over exponentially many subsets S.

The 2026 result can be viewed as an *explicit Hall-to-rank transformation*. From G we construct in NC a polynomial-size numerical matrix B_G such that

$$G \text{ has no perfect matching} \iff \ker B_G \neq \{0\}. \tag{1.2}$$

Since exact matrix rank over $\mathbb{Q}$ is in uniform NC^2, (1.2) gives the desired deterministic parallel algorithm.

This is a useful modern example of a recurring complexity-theoretic phenomenon:

global combinatorial obstruction
$\Downarrow$
redundant algebraic encoding
$\Downarrow$
simple parallel verification

The difficult step is not the final rank computation; linear algebra has been efficiently parallelizable for decades. The difficult step is discovering the right matrix whose rank means exactly what we want.

2 Historical arc: from random genericity to explicit genericity

There are two classical facts to recall.

Edmonds–Lovász. Put an independent indeterminate x_e on every edge and form the symbolic bipartite adjacency matrix. Its determinant is a nonzero polynomial exactly when G has a perfect matching. Lovász's 1979 algorithm substitutes random field elements for the x_e's. A nonzero low-degree polynomial survives a random evaluation with high probability, and the resulting numerical determinant or rank can be computed in NC. Thus bipartite perfect-matching decision is in randomized NC.

The derandomization problem. For decades the randomness resisted complete removal. Fenner–Gurjar–Thierauf obtained quasi-NC, and in 2026 Chatterjee–Ghosh–Gurjar–Raj–Thierauf obtained the first deterministic NC algorithm. Kopparty–Saraf then gave a particularly clean decision criterion, replacing the folded-Wronskian presentation by ordinary multiplicities and an ordinary Wronskian.

The new proof should not be thought of as "finally finding the right deterministic substitution into the old determinant." It changes the algebraic representation itself.

Old algorithm: random evaluation supplies genericity.
New algorithm: build a larger explicit linear system in which genericity is forced structurally.

For the decision-only story, the Isolation Lemma is historically nearby but logically unnecessary. Isolation is central to the classical randomized *search* algorithm; we will not use it below.

3 Proof at a glance

We work over $\mathbb{Q}$ for exposition. Choose $2n$ distinct rational points

$$\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n$$

with polynomial-bit descriptions. We choose three integer scales A, L, K satisfying

$$K > nA + n^2, \tag{1}$$
$$L > A + 2n, \tag{2}$$
$$t(A+1) > (t-1)L \qquad (1 \le t \le n). \tag{3}$$

There is no quantitative subtlety here: these can be arranged with

$$A, L = \Theta(n^2), \qquad L - A = \Theta(n), \qquad K = \Theta(n^3), \tag{3.4}$$

with suitably separated constant factors. Nothing in the proof depends on optimizing them.

Put $D := K + A$. Let $\mathcal{P}_D$ denote the vector space of polynomials of degree at most D.

For every left vertex i, define

$$U_i := \{(X - \alpha_i)^K A_i(X) : \deg A_i \le A\} \subseteq \mathcal{P}_D. \tag{3.5}$$

Thus U_i has dimension $A+1$, and every nonzero element of U_i has a huge, private K-fold zero at α_i.

For every right vertex j, define

$$V_j := \{Q \in \mathcal{P}_D : (X - \beta_j)^L \text{ divides } Q\}. \tag{3.6}$$

Membership in V_j is codimension L: it is equivalent to the L linear equations

$$Q(\beta_j) = Q'(\beta_j) = \cdots = Q^{(L-1)}(\beta_j) = 0. \tag{3.7}$$

Given $P_i \in U_i$, let the polynomial seen by right vertex j be the sum of its neighbors:

$$Q_j := \sum_{i \sim j} P_i. \tag{3.8}$$

We will prove the criterion

$$\boxed{G \text{ has no perfect matching} \iff \exists (P_1, \dots, P_n) \neq 0 : P_i \in U_i \text{ and } Q_j \in V_j \text{ for every } j.} \tag{3.9}$$

The right side is one homogeneous linear system in the coefficients of the A_i's. Hence (3.9) is already almost the algorithm.

The forward implication in (3.9) is pure dimension counting. The reverse implication needs exactly one fact: the high-multiplicity zeros cannot occur accidentally in too small a polynomial space. We prove that next.

4 The one algebraic lemma: a root budget for a polynomial space

The usual root bound says that a nonzero degree-D polynomial has at most D roots, counting multiplicity. We need the same philosophy when we have not one polynomial but an s-dimensional space of them.

For a nonzero polynomial f and a point γ, write $\mathrm{mult}(f, \gamma)$ for the multiplicity of the zero at γ. If $W \subseteq \mathbb{Q}[X]$ is a polynomial space, define

$$N_W(\gamma) := \max_{0 \neq f \in W} \mathrm{mult}(f, \gamma).$$

Lemma 4.1 (Wronskian root budget)**.** *Let $W \subseteq \mathbb{Q}[X]$ be s-dimensional and suppose every polynomial in W has degree at most D. Then*

$$\sum_{\gamma} \max\{N_W(\gamma) - s + 1, 0\} \leq sD, \tag{4.1}$$

where the sum may be taken over all rational or complex points.

Proof. Choose a basis $f_1, \ldots, f_s$ of W and form the Wronskian

$$R(X) := \det \begin{pmatrix} f_1(X) & f_1'(X) & \cdots & f_1^{(s-1)}(X) \\ \vdots & \vdots & & \vdots \\ f_s(X) & f_s'(X) & \cdots & f_s^{(s-1)}(X) \end{pmatrix}. \tag{4.2}$$

Over characteristic zero, linear independence implies $R \not\equiv 0$. For completeness, choose a basis of W with distinct degrees $d_1 < \cdots < d_s$. The leading coefficient of (4.2) is a nonzero scalar times the Vandermonde $\prod_{i<j}(d_j - d_i)$, so it cannot vanish.

Clearly

$$\deg R \leq sD. \tag{4.3}$$

Now fix γ, and choose $f \in W$ having multiplicity $N_W(\gamma)$ there. Change the basis so that f is one of the basis elements. Its rth derivative is divisible by

$$(X - \gamma)^{N_W(\gamma) - r}.$$

Every determinant term therefore contains a factor

$$(X - \gamma)^{\max\{N_W(\gamma) - s + 1, 0\}}.$$

Hence R has at least that multiplicity at γ. Summing over distinct γ and using the ordinary degree-versus-roots bound for the nonzero polynomial R proves (4.1). □

When $s = 1$, Lemma 4.1 is exactly the usual root bound. The Wronskian is merely the device that upgrades "one low-degree polynomial cannot vanish too much" to "one low-dimensional space of low-degree polynomials cannot contain too many highly vanishing vectors."

This is the right conceptual connection to polynomial fingerprinting, Schwartz–Zippel, and sum-check: the same finite-root-budget idea reappears, but one level higher.

5 Private roots force independence

The first use of the root budget is to show that the spaces U_i behave as though they were in generic position.

Lemma 5.1 (Explicit genericity)**.** *Take distinct left vertices $i_1, \dots, i_t$ and choose arbitrary nonzero polynomials*

$$P_{i_r} \in U_{i_r}.$$

Then $P_{i_1}, \dots, P_{i_t}$ are linearly independent.

Proof. Suppose not. Choose a minimal independent subcollection spanning all of them, of size $s < t$, and let

$$W := \operatorname{span}(P_{i_1}, \dots, P_{i_s}).$$

The space W has dimension s and degree at most $D = K + A$.

At each of the s private points belonging to the spanning polynomials, W contains a polynomial with a K-fold zero. If W also contained a nonzero vector from another U_i, it would have such a K-fold zero at one additional private point. Lemma 4.1 would then force

$$(s+1)(K-s+1) \le s(K+A),$$

which rearranges to

$$K \le sA + s^2 - 1 \le nA + n^2.$$

This contradicts (3.1). Thus no omitted P_i can lie in the span, contradicting our choice of the spanning set. □

The lemma is the deterministic substitute for a statement one would normally justify by randomness: independently chosen vectors from sufficiently generic subspaces should not satisfy accidental linear relations. Here their private high-multiplicity zeros make this genericity explicit.

6 The Hall-to-kernel criterion

We now prove (3.9).

6.1 Hall deficiency gives a kernel vector

Suppose G has no perfect matching. By Hall, choose $I \subseteq L$ with

$$|I| = t, \qquad |N(I)| \le t - 1.$$

We set $P_i = 0$ outside I and choose the $P_i \in U_i$ for $i \in I$.

There are

$$t(A+1)$$

free coefficients. Only right vertices in $N(I)$ impose nontrivial conditions. For each such right vertex j, requiring

$$Q_j = \sum_{i \sim j} P_i \in V_j$$

imposes at most L homogeneous linear constraints. Thus the entire system has at most

$$(t-1)L$$

constraints.

By (3.3), the number of unknowns exceeds the number of constraints. Hence there is a nonzero solution. For $j \notin N(I)$ the neighbor sum is identically zero, which already belongs to V_j.

So every Hall-deficient set automatically produces a nontrivial kernel vector.

Hall deficiency has become linear-algebraic deficiency. A set I with $|N(I)| < |I|$ supplies more polynomial degrees of freedom than there are vanishing constraints on its neighborhood.

6.2 A kernel vector gives a Hall obstruction

Conversely, suppose there is a nonzero solution $(P_1, \dots, P_n)$ to the right side of (3.9). Let

$$I := \{i : P_i \neq 0\}, \qquad t := |I|, \qquad J := N(I).$$

By Lemma 5.1, the P_i for $i \in I$ are linearly independent, so

$$W := \operatorname{span}\{P_i : i \in I\} \quad \text{has dimension } t. \tag{6.1}$$

For every $j \in J$, the neighbor sum

$$Q_j = \sum_{i \sim j} P_i$$

is a nonzero element of W: at least one coefficient is 1, and the P_i are independent. Since the linear system requires $Q_j \in V_j$, the space W contains a nonzero polynomial with an L-fold zero at β_j.

Meanwhile, for every $i \in I$, the polynomial $P_i \in W$ has a K-fold zero at its private point α_i.

Apply Lemma 4.1 to the t-dimensional space W. The private left roots and the neighbor roots contribute at least

$$t(K - t + 1) + |J|(L - t + 1) \tag{6.2}$$

to the root budget, while the total budget is at most

$$tD = t(K + A). \tag{6.3}$$

If Hall held for I, then $|J| \geq t$. Combining (6.2) and (6.3) would give

$$K + L - 2t + 2 \leq K + A,$$

or

$$L \leq A + 2t - 2 \leq A + 2n,$$

contradicting (3.2). Therefore

$$|N(I)| < |I|.$$

So the support of every nonzero kernel vector itself reveals a Hall obstruction.

This completes the proof of (3.9).

7 From the criterion to one matrix rank computation

Write

$$P_i(X) = (X - \alpha_i)^K A_i(X), \qquad A_i(X) = \sum_{h=0}^{A} c_{i,h} X^h.$$

The coefficients $c_{i,h}$ are the unknowns. There are

$$n(A+1) = \Theta(n^3)$$

of them.

For every right vertex j and every derivative order $0 \le r < L$, impose

$$\frac{d^r}{dX^r}\left(\sum_{i\sim j}(X-\alpha_i)^K A_i(X)\right)\Bigg|_{X=\beta_j} = 0. \tag{7.1}$$

These are homogeneous linear equations in the $c_{i,h}$. There are

$$nL = \Theta(n^3)$$

equations. Thus (7.1) is represented by a $\Theta(n^3) \times \Theta(n^3)$ matrix B_G whose entries are explicitly

$$(B_G)_{(j,r),(i,h)} = \mathbf{1}_{(i,j)\in E}\,\frac{d^r}{dX^r}\big((X-\alpha_i)^K X^h\big)\Big|_{X=\beta_j}. \tag{7.2}$$

The Hall-to-kernel criterion says precisely

$$G \text{ has no perfect matching} \iff \operatorname{rank}(B_G) < n(A+1). \tag{7.3}$$

Choose the α_i, β_j to be simple distinct integers. Since A, L, K are polynomial in n, every entry in (7.2) has polynomial bit length and can be generated in NC using standard parallel arithmetic. Exact rank of a rational matrix is in uniform NC^2. Therefore:

Theorem 7.1 (CGGRT 2026; decision form via Kopparty–Saraf)**.** *Bipartite perfect-matching decision is in deterministic uniform NC^2.*

The original breakthrough proves more, including search and weighted variants. None of that machinery is needed for the decision theorem above.

8 What is the proof really doing?

There are two complementary ways to remember it.

8.1 Hall expansion versus a finite root budget

A nonzero solution picks a set I of active left vertices. Their private roots ensure that their polynomials form a basis of a $|I|$-dimensional space. Every neighbor of I then forces that same space to contain another polynomial with a large zero at a fresh point. If I had at least $|I|$ neighbors, the Wronskian would accumulate more zero multiplicity than its degree permits.

Thus the proof can be remembered in three short phrases.

Private roots force independence. Neighbor roots measure expansion. The Wronskian has a finite root budget. If Hall expansion held, that budget would be exceeded.

8.2 Random genericity versus explicit genericity

In the classical Lovász algorithm, random substitution is used to avoid accidental algebraic cancellation. In the new construction, each left vertex gets a polynomial subspace U_i arranged so that choosing one nonzero vector from each of several distinct U_i's can never create an accidental dependence. The private roots are an explicit pseudorandomness mechanism.

In the language of the underlying papers, this is a subspace-design phenomenon. For a first lecture, however, that terminology should come *after* the proof. The Wronskian root budget already says exactly what the design is buying us.

9 Connections to earlier themes in the course

1. Schwartz–Zippel and polynomial fingerprinting. Students have repeatedly used

$$\deg p \text{ is small and } p \not\equiv 0 \quad \Longrightarrow \quad p \text{ cannot vanish too often.}$$

The Wronskian lemma is the same idea for a low-dimensional family of polynomials. This is the most important conceptual prerequisite.

2. Linear algebra in NC. Earlier, matrix powering gives $NL \subseteq NC^2$, and determinant, inverse, and rank are paradigmatic parallel computations. Matching shows how much remains after those algorithms are known: the decisive research problem can be finding an algebraic representation whose rank encodes the combinatorics exactly.

3. Derandomization need not mean fixing the random bits. The old RNC algorithm says: use a symbolic determinant, then randomly specialize it. The new deterministic algorithm does not merely supply a clever deterministic specialization. It constructs a different algebraic object where the genericity formerly supplied by randomness is built into the geometry of polynomial subspaces.

4. Isolation as a historical sibling, not a prerequisite. The Isolation Lemma belongs to the classical RNC *search* story and also resonates with Valiant–Vazirani elsewhere in the course. But for decision, leaving isolation out makes the conceptual structure much cleaner: Hall $\to$ roots $\to$ Wronskian $\to$ rank.

5. Encoding a global obstruction into local algebra. Hall's obstruction quantifies over exponentially many subsets. The construction adds redundancy so that all those possible obstructions are simultaneously represented by one linear system. This rhymes with other course transformations in which a large global object becomes amenable to local or algebraic verification, though the mathematical mechanisms are different.

10 A lecture version

For a 70–75 minute lecture, the following division seems realistic.

Time	Content
0–8	Hall's theorem; formulate the goal as an explicit Hall-to-rank lemma. Brief Edmonds–Lovász history.
8–18	Invent the polynomial packets U_i and the right-vertex vanishing spaces V_j. Prove the Hall-deficient direction by counting dimensions.
18–35	Prove the Wronskian root-budget lemma. Begin with the $s = 1$ case: this is just the ordinary root bound.
35–45	First use of the lemma: private roots force independence.
45–58	Second use: a nonzero solution plus Hall expansion exhausts the root budget; conclude a Hall obstruction.
58–65	Write down the explicit matrix B_G and invoke rank in NC^2.
65–75	Historical and conceptual coda: random genericity versus explicit genericity; subspace designs; what the full 2026 result adds.

What to omit. For this lecture there is no reason to discuss weighted matching, search, LP duality, folded Wronskians, optimized finite-field parameters, or the general theory of subspace designs. The decision proof is complete without them.

The final board should be:

Hall deficiency = dimension surplus,

Hall expansion = too many high-multiplicity roots,

Wronskian = the finite root budget.

Therefore Hall's exponentially quantified condition is equivalent to one explicit rank test.

References

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